

Environmental Vulnerability and Respiratory Health in Chicago: A Multi-Scale Analysis of Asthma, COPD, Air Quality, Socioeconomic Conditions, and Summer Heat Exposure

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Abstract:

Urban environmental health risks are unevenly distributed across neighborhoods, and these risks may vary by respiratory health outcome and spatial scale. This paper examines how environmental, socioeconomic, and demographic conditions are associated with asthma and chronic obstructive pulmonary disease (COPD) prevalence across Chicago neighborhoods. Because the available datasets are reported at different geographic resolutions, the study adopts a multi-scale analytical design. At the census block group scale, the analysis examines how social, demographic, and environmental vulnerability indicators are associated with asthma prevalence. At the census tract scale, the study evaluates whether neighborhood-scale summer heat exposure, measured by Landsat-derived maximum land surface temperature (LST), is associated with adult COPD prevalence and whether this association is moderated by racial and ethnic minority share. The analysis draws on the Chicago Department of Public Health's Air Quality and Health Index, CDC PLACES small-area health estimates, and Landsat-derived surface temperature data. By combining block group-level vulnerability indicators with tract-level heat-health modeling, this paper contributes to urban environmental health and environmental justice research by showing how respiratory health risks are structured through overlapping environmental exposures, socioeconomic conditions, demographic composition, and spatial scale.

Introduction:

Urban environmental health risks are shaped by the spatial concentration of air pollution, heat exposure, socioeconomic disadvantage, and demographic vulnerability. These risks are particularly important for respiratory health because air pollution and temperature-related stressors can aggravate existing respiratory conditions and contribute to uneven health burdens across urban neighborhoods. Ambient air pollution has been identified as a major environmental health risk, with fine particulate matter and ozone linked to cardiovascular and respiratory disease, including asthma and chronic obstructive pulmonary disease (COPD) (World Health Organization [WHO], 2024; U.S. Environmental Protection Agency [EPA], 2026a, 2026b). The EPA notes that ozone exposure can worsen asthma and bronchitis symptoms, while particle pollution poses elevated risks for people with asthma and COPD (EPA, 2026a, 2026b). These relationships make respiratory health outcomes a useful lens for examining how neighborhood environmental burden and social vulnerability intersect.

Urban heat is another important dimension of environmental health inequality. Land surface temperature (LST) derived from satellite imagery is widely used to characterize spatial variation in urban heat because it captures differences in surface heating across land cover, built form, and neighborhood environments. Landsat surface temperature products are specifically designed to support analyses of surface thermal conditions, including studies of urban heat islands and extreme heat exposure (U.S. Geological Survey [USGS], n.d.). Prior research has shown that heat exposure is not evenly distributed across urban populations. For example, Hsu et al. (2021) found that, across 175 large urbanized areas in the continental United States, the average person of color lived in a census tract with higher summer daytime surface urban heat island intensity than non-Hispanic White residents in nearly all studied urbanized areas. Similarly, Hoffman et al. (2020) found that historically redlined neighborhoods in many U.S. cities experience higher present-day land surface temperatures, suggesting that contemporary heat exposure is partly shaped by long-term patterns of racialized urban development.

Chicago is a useful case for examining these relationships because the city contains substantial neighborhood-level variation in air pollution exposure, socioeconomic vulnerability, demographic composition, and built-environment conditions. The Chicago Department of Public Health's Air Quality and Health Index provides index scores and ranks for each census block group, allowing environmental and population vulnerability to be examined at a fine spatial scale (Chicago Department of Public Health [CDPH], n.d.). The associated indicator documentation identifies block group-level indicators related to environmental burden, population characteristics, and health vulnerability, making the dataset suitable for analyzing spatial patterns of neighborhood environmental health risk (CDPH, n.d.).

This paper therefore, adopts a multi-scale analytical design. The use of both census block groups and census tracts is not an arbitrary mixing of spatial units, but rather reflects the geographic resolution at which the available data are measured and analytically comparable. The block group scale is used where the AQH data provide fine-grained measures of air quality burden, social vulnerability, and population characteristics. The census tract scale is used where Landsat-derived LST summaries and CDC PLACES respiratory health estimates can be harmonized into a common modeling unit. This approach allows the analysis to examine respiratory health vulnerability from two complementary perspectives: first, how social, demographic, and environmental vulnerability indicators are associated with asthma prevalence at the census block group scale; and second, how summer heat exposure is associated with COPD prevalence at the census tract scale, with attention to whether that association differs by racial and ethnic minority composition.

This paper contributes to urban planning and environmental health research in three ways. First, it links air quality burden, social vulnerability, and respiratory health outcomes in a Chicago-specific context. Second, it incorporates remote-sensing-based LST as a neighborhood-scale measure of summer heat exposure. Third, it explicitly addresses spatial scale by using block group and census tract analyses according to the geographic resolution of the underlying datasets. This multi-scale approach avoids direct comparison of coefficients across incompatible spatial units while still allowing the findings to inform a broader interpretation of environmental health vulnerability across Chicago neighborhoods.

Keywords:

Environmental health; Respiratory health; Urban heat; Air quality; Socioeconomic vulnerability; Environmental justice

Literature Review:

A study conducted by Douglas et al. (2019) on asthma emergency department visits (AEDVs) in Los Angeles County investigates how environmental exposures, like diesel particulate matter (DPM) and open spaces, relate to outcomes in asthma prevalence. The study attempts to address the broader issue of spatial health disparities that are linked to pollution and lack of access to green spaces. The study utilizes census tract-level data and applies an Ordinary Least Squares (OLS) regression model to estimate the global associations. It also controls for different sociodemographic factors, such as age, poverty, unemployment, race, ethnicity, etc. The OLS model demonstrates that DPM is positively associated with AEDVs, while PPOS is negatively associated, indicating a potential protective effect of greenspace. The study also employs Geographically Weighted Regression (GWR) because a limitation was cited in the OLS as it assumed constant relationships across space. The model was improved with GWR. It improved the model fit and captured better spatial variation in relationships. Ultimately, the study found that natural park visits are associated with lower asthma risks, but the relationship is non-linear. The findings conclude that while OLS is useful for identifying overall trends, it fails to account for spatial heterogeneity. However, GWR provided a more localized approach to ensure accurate findings. The study concluded with an understanding of how social and environmental factors influence asthma outcomes across different neighborhoods in Chicago.

In a 2026 longitudinal multistate transition analysis of the COPDGene cohort, Sanchez-Romero et al. examined movement across COPD severity stages and found that regression is a notable feature of early disease. Individuals with mild COPD (GOLD 1) frequently transitioned not only to more severe stages but also regressed to less severe or normal spirometric states. In contrast, the regression signal disappears in advanced disease (GOLD 3–4), where transitions are dominated by further decline and mortality. The study concludes that early COPD is highly dynamic and may include reversible components or measurement variability, highlighting a potential window for early intervention and challenging the assumption that COPD progression is strictly one-directional.

Respiratory health data, however, are often available or more analytically stable at a different spatial resolution. CDC PLACES provides model-based small-area estimates of chronic disease and health-related measures at multiple local geographies, including census tracts and ZIP Code Tabulation Areas (Centers for Disease Control and Prevention [CDC], 2025). The PLACES methodology uses multilevel regression and poststratification to generate local estimates for adults aged 18 years and older (CDC, 2025). Greenlund et al. (2022) describe PLACES as a platform that provides standardized local-level chronic disease estimates for counties, places, census tracts, and ZIP Code Tabulation Areas, supporting public health planning where directly observed local health data are limited. Because these health outcomes are model-based estimates rather than direct clinical observations, they should be interpreted as small-area indicators of population health burden rather than individual-level disease records.

The current research states that there are high disparities in asthma prevalence in communities across the United States and, more specifically, ethnic minorities and other vulnerable groups.

The center for Disease Control and Prevention, explains that demographics determine your health outcomes. Adding on, the World Health Organization states that social factors such as education, language barriers, and the age pyramid or age structure impact variation in health conditions. Likewise, the American Lung Association highlights that neighborhoods with limited resources, as well as low access to social services, experience substantially worse respiratory prognosis. To conclude, the research states that it's not purely biological but rather a mix of personal decisions and social factors that determines one's state of being amongst contracting a respiratory disease.

Research questions:

The main research question guiding this paper is:

How are environmental, socioeconomic, and demographic conditions associated with respiratory health outcomes across Chicago neighborhoods, and how do these associations vary across block group and census tract scales?

Three sub-questions follow from this broader question:

1. *At the census block group scale, how are social, demographic, and environmental vulnerability indicators associated with asthma prevalence across Chicago neighborhoods?*
2. *How does neighborhood-scale summer heat exposure, measured by Landsat-derived maximum land surface temperature, relate to adult COPD prevalence across Chicago census tracts, and does this relationship differ by racial and ethnic minority composition?*
3. *To what extent do environmental exposure clusters reflect broader patterns of cumulative environmental disadvantages across Chicago communities?*

Data:

The data used in this analysis comes from the Chicago Department of Public Health Air Quality and Health Index dataset. This dataset was developed as part of the City of Chicago's broader environmental health initiative under Healthy Chicago 2025 to identify communities disproportionately affected by air pollution and related health risks. This dataset compiles community-level indicators on air quality, health outcomes, and social and demographic characteristics from 2020. It is primarily organized at the census block group level, allowing for fine-grained geographic analysis across neighborhoods within Chicago. This assists in providing insight into the spatial inequalities that exist in environmental exposure and health outcomes

across Chicago. There are 2120 rows and 107 variables in the original dataset. There were no missing values in the dataset. The dataset was cleaned to ensure there were 54 variables utilized for this research. The dataset was cleaned to include the following variables:

Category	Variables_Used
Health Outcomes	asthma_perc, copd_perc, chd_perc, low_birthweight_perc
Demographic Characteristics	minority_pct, low_income_pct, less_hs_pct, linguistic_iso_pct, under5_pct, over64_pct, population
Environmental Exposure	proximity_traffic, proximity_national_sites, proximity_risk_sites, proximity_waste_sites
Air Pollution Indicators	pm25_num, ozone_num, diesel_pm_num, cancer_risk_num, respiratory_hazard_num
Socioeconomic Conditions	unemp_rate, housing_burden_perc, total_births_perc

The final set of variables used in this analysis were selected to reflect key dimensions of environmental health vulnerability. Together, these variables capture both the social and environmental determinants of respiratory health across Chicago neighborhoods.

Summary Statistics

Summary Statistics: Asthma, COPD, and Environmental Indicators							
	Unique	Missing Pct.	Mean	SD	Min	Median	Max
asthma_perc	77	0	9.4	1.9	6.7	8.5	14.9
copd_perc	116	0	6.5	2.5	1.7	6.2	17.4
minority_pct	1802	0	0.7	0.3	0.0	0.8	1.0
low_income_pct	2096	0	0.4	0.2	0.0	0.4	1.0
less_hs_pct	1923	0	0.2	0.1	0.0	0.1	0.7
linguistic_iso_pct	1283	0	0.1	0.1	0.0	0.0	0.6
under5_pct	1875	0	0.1	0.0	0.0	0.1	0.3
over64_pct	2064	0	0.1	0.1	0.0	0.1	0.8
proximity_traffic	2116	0	1418.7	2962.6	0.0	421.7	34262.4
proximity_risk_sites	2116	0	1.9	1.9	0.0	1.2	17.4
pm25_num	779	0	9.8	0.1	9.4	9.8	10.1
ozone_num	779	0	45.4	0.3	44.5	45.4	46.3
diesel_pm_num	779	0	1.2	0.5	0.7	1.0	3.2

The summary statistics indicate substantial variation in the variables for the variables across Chicago neighborhoods. The mean for asthma prevalence is 9.4% while COPD prevalence is lower at 6.5% but exhibits greater dispersion. Social variables show high variability, with the minority population averaging 70%. Low-income and educational attainment indicators vary widely across census block groups.

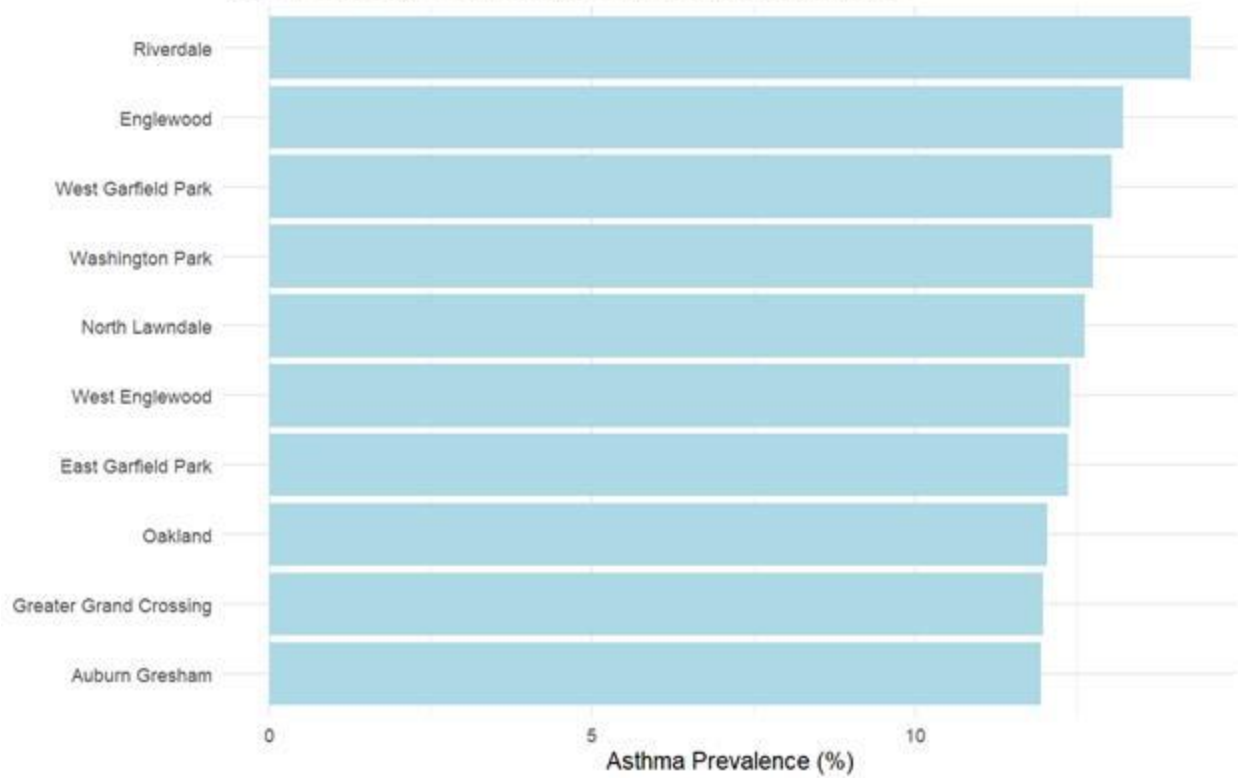
Correlation Table

Variable	asthma_perc	copd_perc	minority_pct	low_income_pct	less_hs_pct	pm25_num	ozone_num	proximity_traffic
asthma_perc	1.00	0.88	0.71	0.61	0.20	0.31	0.04	-0.15
copd_perc	0.88	1.00	0.68	0.64	0.36	0.37	-0.05	-0.16
minority_pct	0.71	0.68	1.00	0.72	0.56	0.46	-0.06	-0.14
low_income_pct	0.61	0.64	0.72	1.00	0.59	0.29	-0.04	-0.13
less_hs_pct	0.20	0.36	0.56	0.59	1.00	0.44	-0.27	-0.14
pm25_num	0.31	0.37	0.46	0.29	0.44	1.00	-0.68	-0.06
ozone_num	0.04	-0.05	-0.06	-0.04	-0.27	-0.68	1.00	-0.08
proximity_traffic	-0.15	-0.16	-0.14	-0.13	-0.14	-0.06	-0.08	1.00

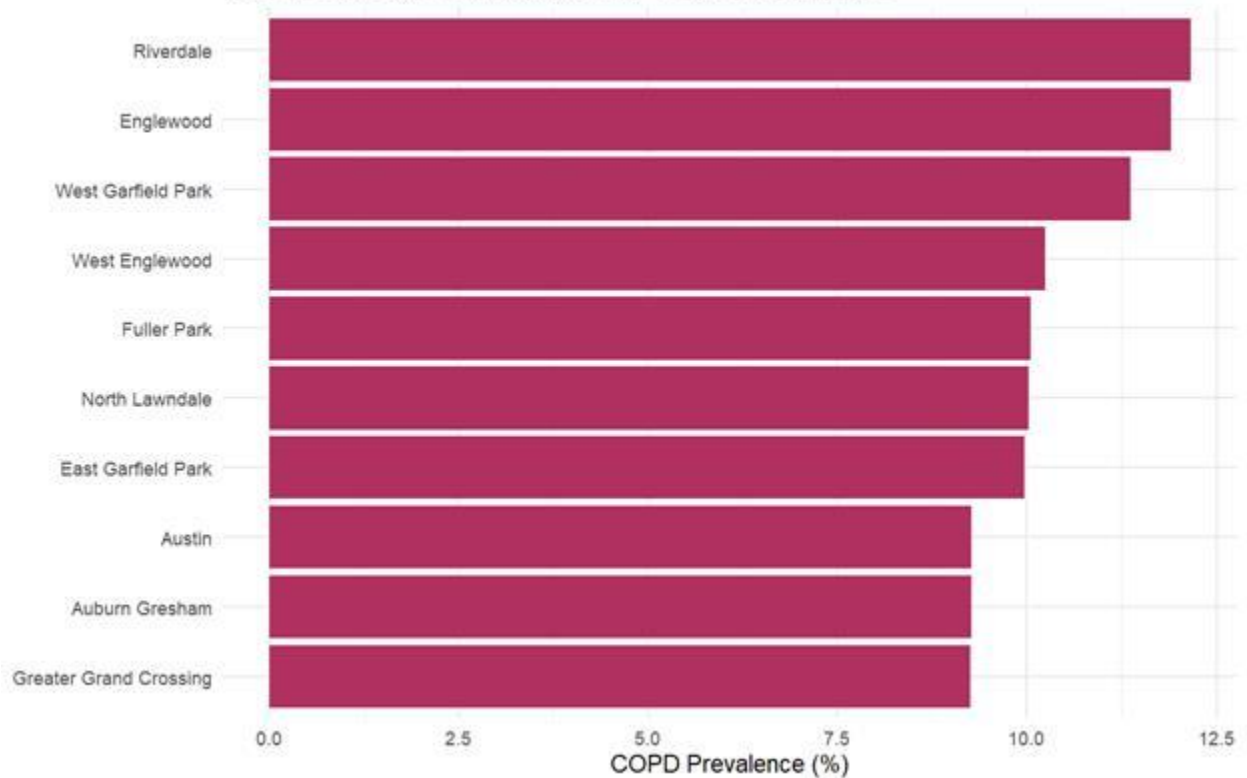
The correlation matrix reveals several strong relationships each of the variables. Asthma and COPD prevalence are highly correlated (corr = 0.88). They are also strongly associated with the share of minority residents (asthma_perc: corr = 0.71; copdperc: corr = 0.68) and moderately associated with low-income population (asthma_perc: corr = 0.61; COPD: corrr = 0.64), indicating that socially disadvantaged neighborhoods bear a greater burden of respiratory disease. PM2.5 exhibits modest positive correlations with both asthma (r = 0.31) and COPD (r = 0.37), suggesting some association with respiratory outcomes. Notably, PM2.5 and ozone are strongly negatively correlated (r = -0.68). On the other hand, proximity to traffic is weakly negatively correlated with the variables.

Top 10 Chicago Communities by Asthma and COPD Prevalence

Top 10 Chicago Communities by Asthma Prevalence



Top 10 Chicago Communities by COPD Prevalence



To examine the spatial distribution of respiratory health burdens across Chicago, the analysis identified the ten community areas with the highest average asthma prevalence and the ten community areas with the highest average COPD prevalence. Interestingly, Riverdale, Englewood, and West Garfield Park were the top three community areas with the highest percentage of asthma prevalence and COPD prevalence. The visualization was created using ggplot().

Methods & Results:

Model 1: Asthma analysis

The dependent variable in this model is `asthma_perc` and it measures the crude prevalence of asthma across all block groups in the City of Chicago. The independent variables signify important social as well as demographic attributes. Moreover, `minority_pct` tells us that the percentage of the population in Chicago that classifies themselves as a minority. The variable `less_hs_pct` is the percentage of individuals who claim they do not have a high school diploma and helps to measure the education levels and attainment within the community.

`linguistic_iso_pct` measures the percent of the population that does not speak English fluently, which reflects the hurdles of language barrier. `over64_pct` is percent for the population who are 64 and older. This variable captures the group that is most vulnerable socially given health deteriorates and can capture insights on social disparities. The hypothesis states that communities that have less resources and more vulnerabilities, such as language barriers, less education have higher asthma rates due to their position in society.

Why the Log-Lin Model was chosen:

The model chosen to conduct would be the log to linear multiple regression model. It is also important to note that the dependent variable `asthma DCRP` is a rate that is more likely to be skewed. When we implement the logarithm into the multiple regression model, it helps to stabilize the distribution as well as help diminish extreme values or otherwise known as outliers. Additionally, these methods allow for model output tables to be interpreted in percentage terms which are a great way of displaying meaningful public health context. In comparison to other models such as a linear model in levels, the logarithmic transformation offers a greater functioning form when it comes to modeling proportional outputs as public health changes.



The plot shows residuals on the y-axis (ranging from -0.2 to 0.4) and standardized residuals on the x-axis (ranging from 0.0 to 1.5). A horizontal line is drawn at y=0. Most data points are clustered around the zero line, with a few outliers at higher standardized residual values (around 1.5).

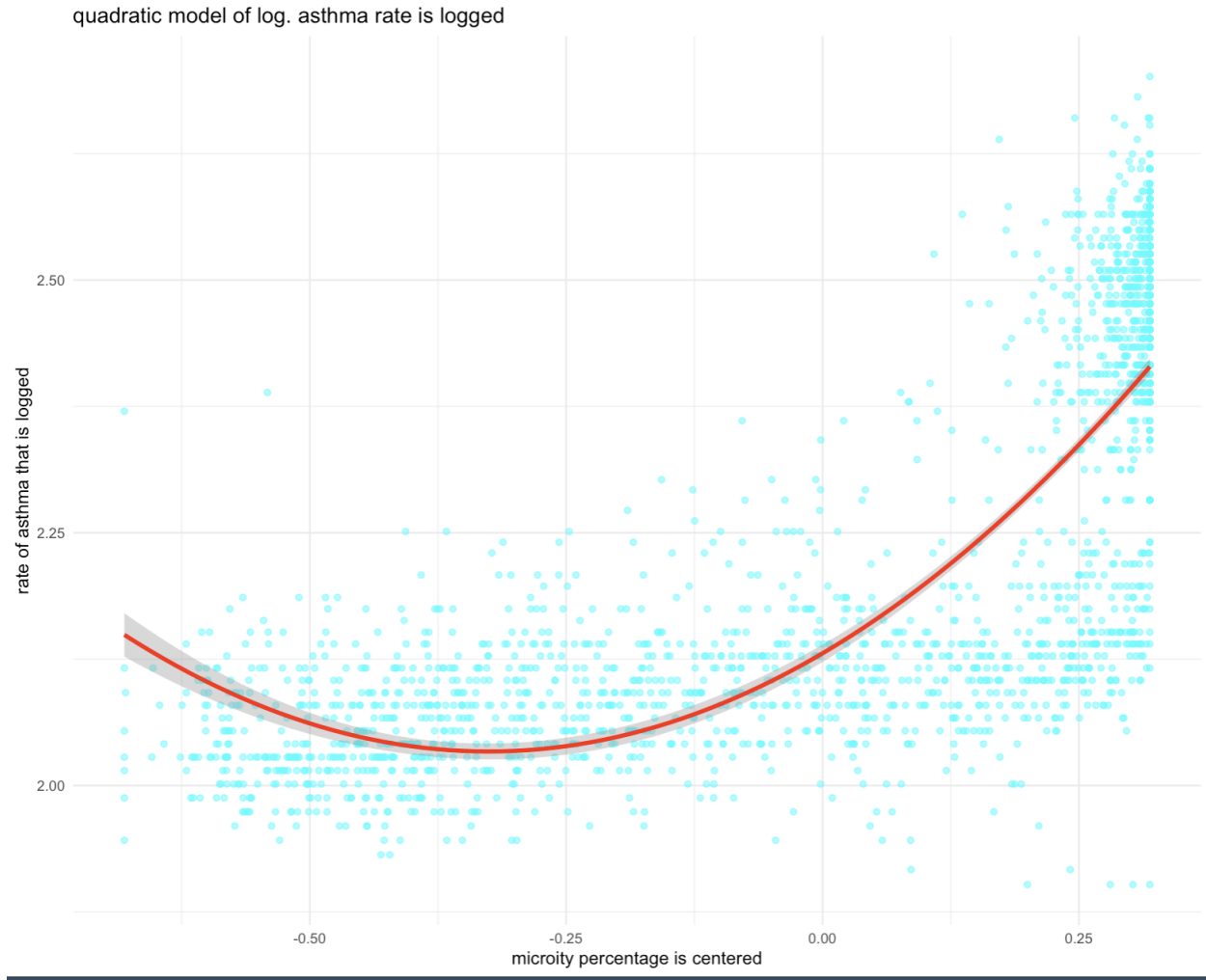
	quadratic model	interaction model	standardized model
(Intercept)	2.158*** (0.004)	2.218*** (0.003)	2.221*** (0.002)
minority_pct_c	0.577*** (0.010)	0.474*** (0.014)	
I(minority_pct_c^2)	0.638*** (0.031)		
less_hs_pct_c	-0.049+ (0.028)	-0.018 (0.036)	
linguistic_iso_pct_c	-0.581*** (0.035)	-0.725*** (0.034)	
over64_pct_c	0.157*** (0.030)	0.215*** (0.031)	
minority_pct_c × less_hs_pct_c		0.139 (0.099)	
minority_pct_z			0.145*** (0.003)
less_hs_pct_z			0.002 (0.004)
linguistic_iso_pct_z			-0.077*** (0.004)
over64_pct_z			0.018*** (0.003)
Num.Obs.	2120	2120	2120
R2	0.747	0.700	0.699
R2 Adj.	0.747	0.699	0.699
AIC	5546.1	5910.0	5910.5
BIC	5585.7	5949.7	5944.5
Log.Lik.	1942.376	1760.412	1759.176
F	1481.747	1029.324	1244.668
RMSE	0.10	0.11	0.11
Std.Errors	HC1	HC1	HC1

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

The regression models inform us that there are multiple social as well as demographic factors that are significantly associated with the asthma prevalence variable. When we look at the model that has the quadratic term, the coefficient that is about minority population or (minority_pct_c) results in a positive sign, along with being statistically significant ($B = 0.577$, $p < 0.01$). Moving forward, holding other variables constant, a one unit increase in the percentage of minority population is associated with a higher asthma rate overall. The quadratic term for the minority population ($B = 0.638$, $p < 0.01$) also has a positive sign as well as being statistically significant like the previous variable. This clears up the doubt that there is a nonlinear relationship as the effect becomes larger at higher labels in the minority population. The variable of the percentage of folks who do not speak English enough to understand it (linguistic_iso_pct_c) is negative but highly significant ($B = -0.581$, $p < 0.01$). On the other hand, the percentage of the population that

is 64 years or older (`over64_pct_c`) has a positive sign and is highly statistically significant ($B = 0.157$, $P < 0.01$). The variable on education (`less_hs_pct_c`) is significantly weak when it comes to the quadratic model and is statistically insignificant in the model with the interaction term. Undoubtedly, there is a lack of evidence of a strong independent impact. When we look at the model with the interaction term present, the minority population in between education ($B = 0.139$) shows no statistical significance and advice that the impact of minority population on the dependent variable or asthma prevalence is not meaningful across levels of educational attainment. When we compared the models based on fit, the model with the quadratic term present is much better than the model with the interaction term present. With the R-squared that is higher at $-.747$ in comparison to the r squared for the interaction term 0.700 . The lower AIC and BIC values and a much lower RMSE respectively. This provides a general summary that it is a better representation of demographic attributes with asthma prevalence than when it comes to including an interaction term and its effects. The standardized model was aggregated to make comparisons on the independent variables within the same scale. More specifically, we know that the variables in the model are in different percentages as well as units which can be tricky but, standardization helps to provide a more holistic and meaningful resemblance across multiple predictors at hand. When we turn variables into standardized z cores, these coefficients in the model exhibit the power of association each of the coefficients have with the asthma prevalence rate. This directly makes an improvement due to the fact that larger coefficients have a much stronger relationship among the dependent variable in comparison to other smaller coefficients. Lastly, it strengthened our analysis by improving the way we differentiate variable comparability among social and demographic variables in the regression model.

Diagnostics Plots were built in order to access the model assumptions. The visualization demonstrates that there is minor nonlinearity and exists due to the inclusion of the quadratic term into our model. Very evidently so, the Q-Q plot tells us that residuals are approximately normally skewed distributed. Additionally, the scale of the location plot lets us know that there is heteroskedasticity present and is taken care of using robust standard errors. Taking all things into consideration, there seems to be no presence of violations of the model assumptions.



Model 2: Chronic Obstructive Pulmonary Disease (COPD) analysis

$$COPDi = \beta_0 + \beta_1 LSTmaxZi + \beta_2 Minorityi + \beta_3 (LSTmaxZi \times Minorityi) + \gamma Xi + \epsilon_i$$

The final OLS interaction model examines whether the association between summer maximum land surface temperature (LST) and adult COPD prevalence varies by racial and ethnic minority share across Chicago census tracts. The dependent variable is adult COPD prevalence, while the primary explanatory variable is standardized summer maximum LST. Minority share is included both as a main effect and as an interaction term with LST. The model further adjusts for demographic vulnerability, socioeconomic conditions, air pollution exposure, diesel particulate

matter, and traffic-related exposure. The final specification is therefore designed to test not only whether hotter census tracts have higher COPD prevalence, but also whether this association differs across tracts with different racial and ethnic compositions.

```
Call:
lm(formula = copd ~ LST_max_z * minor + over64 + under5 + lingiso +
    lowinc + lesshs + unemp + hsgburden + pm25 + ozone + dslpm +
    log_ptraf, data = df)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-3.5668 -0.6904 -0.0317  0.5989 11.2845
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  49.045392  16.563853   2.961  0.00316 **
LST_max_z     0.378494   0.117724   3.215  0.00136 **
minor        -0.002016   0.002964  -0.680  0.49652
over64        0.140207   0.007236  19.376 < 2e-16 ***
under5        0.047687   0.015479   3.081  0.00214 **
lingiso       -0.080930   0.006692 -12.094 < 2e-16 ***
lowinc        0.066527   0.004647  14.315 < 2e-16 ***
lesshs       0.037240   0.006668   5.585 3.26e-08 ***
unemp         0.040230   0.007240   5.557 3.81e-08 ***
hsgburden     0.013363   0.005338   2.503  0.01252 *
pm25          0.125607   0.591597   0.212  0.83192
ozone        -1.069225   0.258051  -4.143 3.81e-05 ***
dslpm        -0.773992   0.091481  -8.461 < 2e-16 ***
log_ptraf     0.055938   0.031516   1.775  0.07631 .
LST_max_z:minor -0.004582   0.001571  -2.917  0.00363 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.128 on 760 degrees of freedom
Multiple R-squared:  0.8263,    Adjusted R-squared:  0.8231
F-statistic: 258.2 on 14 and 760 DF,  p-value: < 2.2e-16
```

The model shows strong overall explanatory power. The multiple R-squared is 0.8263, and the adjusted R-squared is 0.8231, indicating that the included predictors explain a substantial share of cross-tract variation in estimated COPD prevalence. The overall model is statistically significant, with an F-statistic of 258.2 and a p-value below 2.2e-16. However, this high explanatory power should be interpreted cautiously. The COPD outcome is based on small-area health estimates rather than directly observed clinical records, and many predictors capture overlapping demographic, socioeconomic, environmental, and spatial patterns. Therefore, the model fit indicates strong statistical association, but it should not be interpreted as evidence of causal effects.

```
===== 稳健标准误 (HC3) 下的推断 =====
```

```
t test of coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  49.0453924  21.6596778   2.2644 0.0238327 *
LST_max_z     0.3784940  0.1005864   3.7629 0.0001809 ***
minor        -0.0020164  0.0058715  -0.3434 0.7313807
over64        0.1402070  0.0104580  13.4066 < 2.2e-16 ***
under5        0.0476875  0.0185813   2.5664 0.0104658 *
lingiso       -0.0809297  0.0077977 -10.3787 < 2.2e-16 ***
lowinc        0.0665270  0.0086808   7.6637 5.516e-14 ***
lesshs       0.0372397  0.0086631   4.2987 1.942e-05 ***
unemp         0.0402299  0.0110730   3.6331 0.0002988 ***
hsgburden     0.0133631  0.0071089   1.8798 0.0605232 .
pm25          0.1256068  0.7679737   0.1636 0.8701241
ozone        -1.0692247  0.3293005  -3.2470 0.0012176 **
dslpm        -0.7739923  0.0787481  -9.8287 < 2.2e-16 ***
log_ptraf     0.0559381  0.0333605   1.6768 0.0939981 .
LST_max_z:minor -0.0045822  0.0015788  -2.9024 0.0038106 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The coefficient for standardized maximum LST is positive and statistically significant. In the conventional OLS output, the coefficient for LST_max_z is 0.3785 with a p-value of 0.0014. Under HC3 robust standard errors, the coefficient remains 0.3785 and is still statistically significant, with a p-value of 0.00018. This result indicates that, when minority share is set to zero in the model, a one-standard-deviation increase in maximum summer LST is associated with an approximately 0.38 percentage-point increase in COPD prevalence. Because this is an interaction model, this coefficient is conditional on $\text{minor} = 0$. Therefore, it should not be interpreted as the average LST effect across all tracts, especially if census tracts with zero minority population are rare or outside the main empirical range of the data.

The interaction between LST and minority share is negative and statistically significant. The coefficient for LST_max_z:minor is -0.00458 in both the OLS and HC3 robust results, with p-values of 0.0036 and 0.0038, respectively. This means that each one-percentage-point increase in minority share is associated with a 0.00458 percentage-point reduction in the marginal association between LST and COPD prevalence. Substantively, the positive association between LST and COPD is strongest in tracts with lower minority share and becomes progressively weaker as minority share increases. This result does not support the initial environmental justice hypothesis that the LST-COPD association would be stronger in higher-minority tracts. Instead, the model suggests a negative moderation pattern.

The estimated marginal effects further clarify this pattern. When minority share is 20%, a one-standard-deviation increase in maximum LST is associated with a 0.2869 percentage-point increase in COPD prevalence. At 40% minority share, the marginal association declines to 0.1952 percentage points. At 60%, it falls to 0.1036 percentage points. At 80%, the estimated marginal association is only 0.0119 percentage points, which is substantively close to zero. These results indicate that the LST-COPD association is not constant across Chicago census tracts. Rather, it is concentrated in tracts with relatively lower minority share and becomes much weaker in higher-minority tracts.

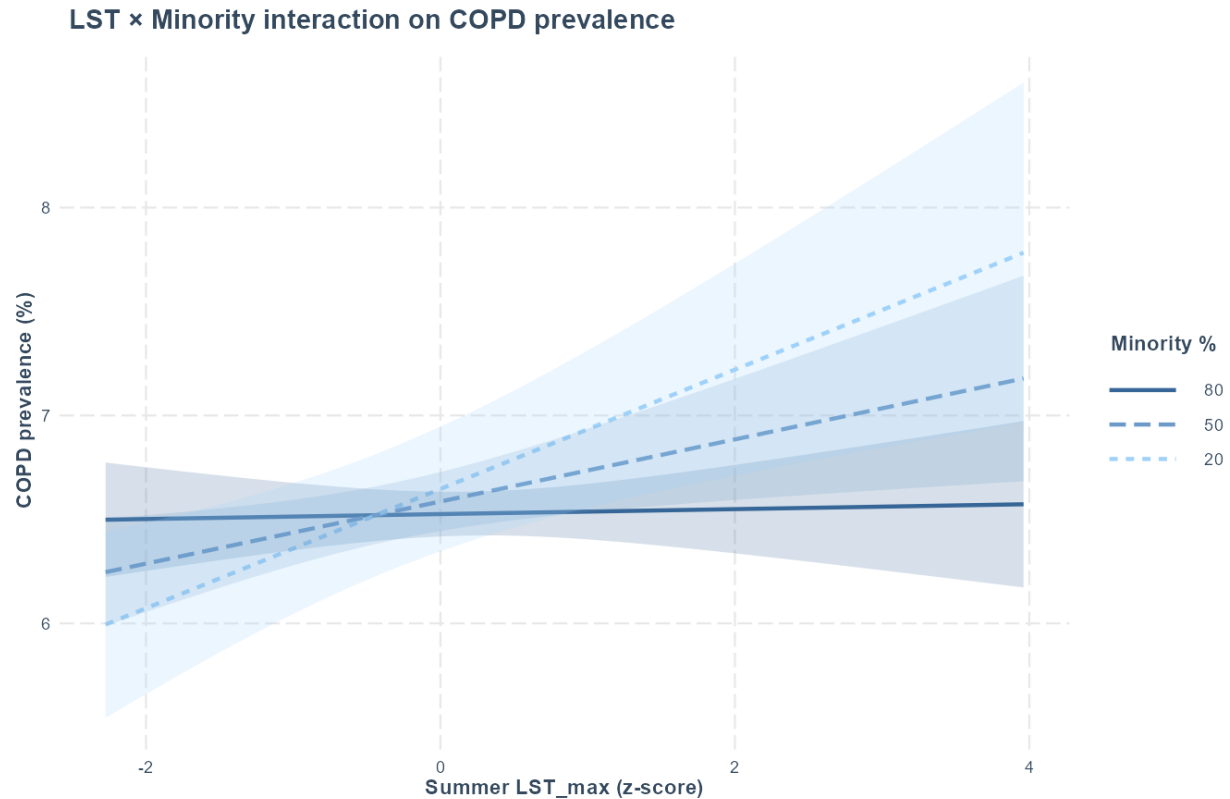


Figure 1 visualizes this interaction by plotting predicted COPD prevalence across standardized maximum LST values at three levels of minority share: 20%, 50%, and 80%. The line for 20% minority share has the steepest positive slope, indicating that higher LST is associated with a clear increase in predicted COPD prevalence in lower-minority tracts. The line for 50% minority share is still positive but less steep. The line for 80% minority share is nearly flat, suggesting little meaningful association between LST and COPD prevalence in high-minority tracts. This visual evidence is consistent with the negative and statistically significant interaction coefficient. The widening confidence intervals at the upper end of the LST distribution also indicate greater uncertainty in predicted COPD prevalence at high LST values.

The Johnson–Neyman analysis identifies the range of minority share values for which the marginal association between LST and COPD is statistically significant. The estimated lower bound is approximately 61.13, while the upper bound is 127.91. Because minority share cannot meaningfully exceed 100%, the substantively relevant threshold is the lower bound. Within the observed range of the data, the LST–COPD association is statistically significant when minority share is below approximately 61%. Above this threshold, the marginal association between LST and COPD is no longer statistically distinguishable from zero.

Johnson-Neyman: where is LST-COPD relationship significant?

Cyan = significant; pink = not significant

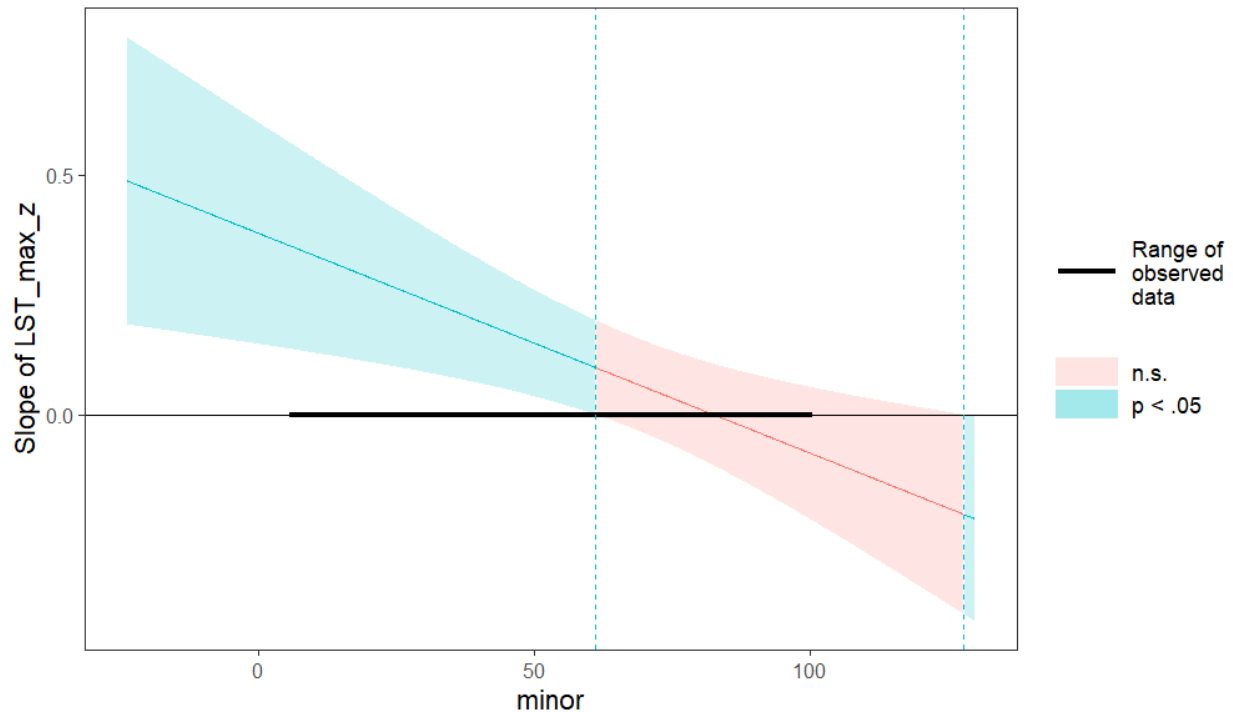
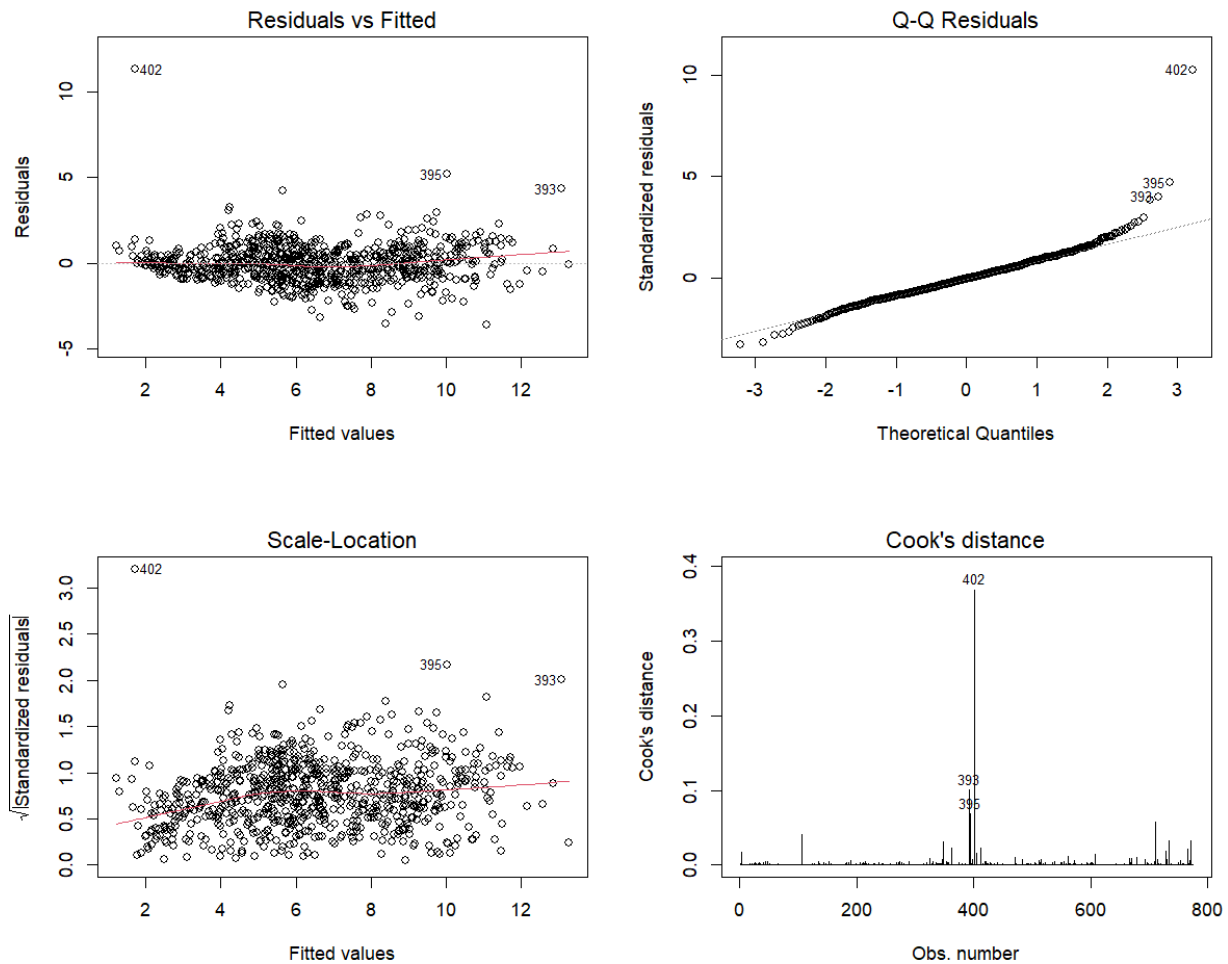


Figure 2 presents the Johnson–Neyman region of significance. The y-axis shows the conditional slope of LST, and the x-axis shows minority share. The cyan region indicates where the conditional LST association is statistically significant, while the pink region indicates where it is not significant. The figure confirms that the LST-COPD association is statistically significant at lower levels of minority share but becomes non-significant after minority share reaches approximately 61%. The upper Johnson–Neyman bound lies outside the meaningful range of minority share and should not be substantively interpreted. The black horizontal line indicates the range of observed data, reinforcing that interpretation should be limited to the empirical range.



The model diagnostics suggest that the results should be interpreted with caution. In Figure 3, the residuals-versus-fitted plot shows that most residuals are centered around zero, but the smoothed trend line is not perfectly flat. This suggests that some functional-form misspecification or omitted nonlinear structure may remain. The Q-Q plot shows that the central part of the residual distribution follows the theoretical normal line reasonably well, but the upper tail deviates substantially. Observation 402 is especially notable as an extreme positive residual. The scale-location plot indicates non-constant residual variance, which is consistent with the Breusch-Pagan test result. The Breusch-Pagan test is statistically significant, with $p = 0.00362$, indicating heteroskedasticity. The use of HC3 robust standard errors is therefore appropriate for statistical inference.

The Cook's distance plot also identifies observation 402 as a highly influential case. Its Cook's distance is much larger than that of the rest of the sample, while observations 393 and 395 also show elevated influence. This suggests that the estimated coefficients may be partly sensitive to a small number of influential census tracts. A sensitivity analysis excluding observation 402, or excluding observations with high Cook's distance, would be useful to evaluate whether the LST

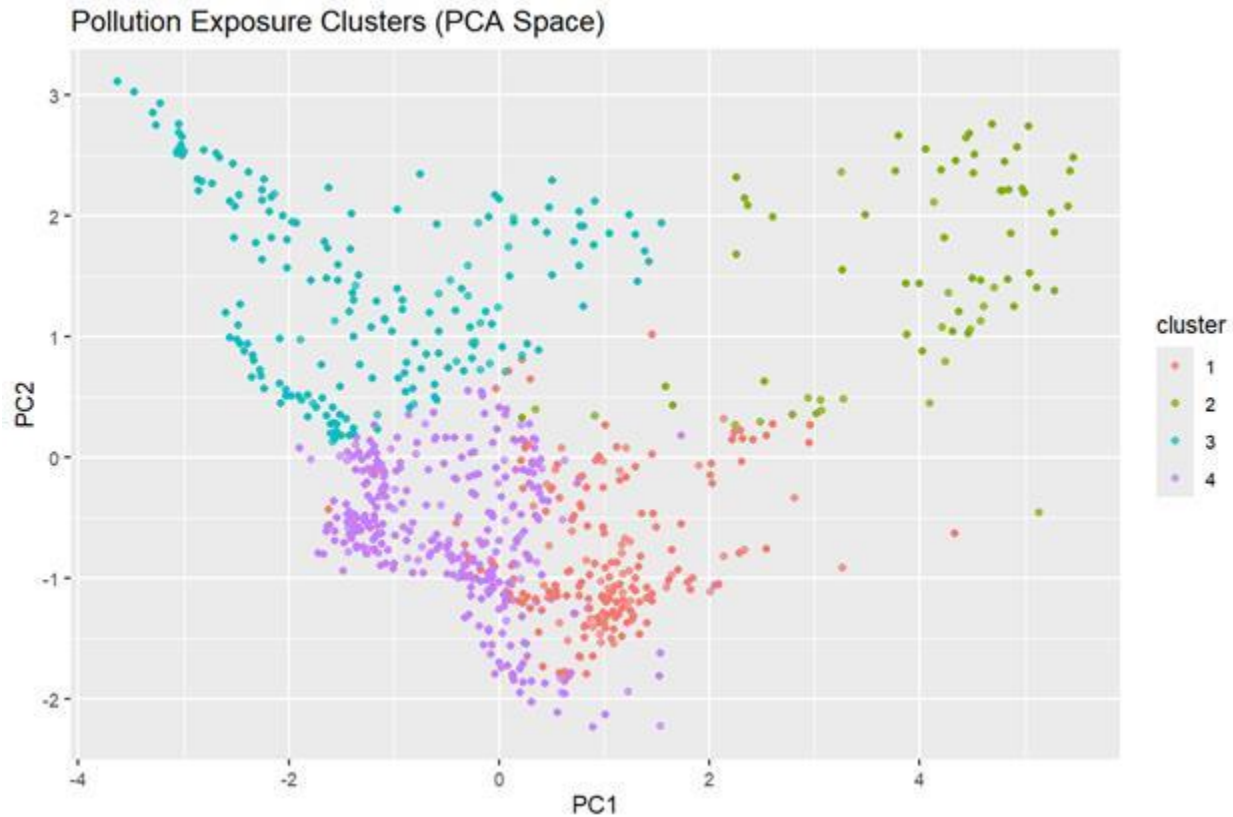
main effect and the LST-minority interaction remain stable. If the interaction remains negative and statistically significant after removing influential cases, the finding would be more robust. If it changes substantially, the result should be reported as sensitive to influential observations.

The formal diagnostic tests reinforce these concerns. The RESET test is statistically significant, with $p = 5.54e-09$, indicating that the current linear specification may not fully capture the underlying structure of the data. This does not necessarily mean that the LST relationship itself is incorrectly specified; rather, it may reflect omitted variables, unmeasured nonlinearities, measurement error, or spatial dependence. The VIF/GVIF results do not indicate severe multicollinearity under the common threshold of 10, but they do show moderate collinearity for several predictors. The largest GVIF values are 6.30 for LST_max_z, 6.30 for minor, 5.33 for lowinc, 4.15 for lesshs, 3.94 for pm25, and 3.28 for ozone. These values suggest that LST, minority share, socioeconomic disadvantage, and pollution exposure share overlapping spatial patterns. As a result, individual coefficients should be interpreted as conditional associations rather than independent causal mechanisms.

Overall, the final model provides evidence that summer maximum LST is positively associated with adult COPD prevalence across Chicago census tracts, but only under certain demographic conditions. The association is strongest in tracts with lower minority share and weakens as minority share increases. The Johnson–Neyman result indicates that the LST-COPD association is statistically significant only when minority share is below approximately 61%. This finding contradicts the initial expectation that higher-minority tracts would show stronger heat-related health vulnerability. A more cautious interpretation is that, in higher-minority or more structurally disadvantaged tracts, COPD prevalence may already be strongly shaped by other social, environmental, and health-related factors, leaving less additional variation to be explained by LST. Possible explanations include baseline saturation, exposure saturation, covariate absorption, spatial dependence, and the model-based nature of the COPD estimates.

Therefore, the central finding should be reported as an OLS-framework conditional association rather than a causal effect. The results suggest that higher summer maximum LST is associated with higher COPD prevalence in lower-minority census tracts, but this association becomes weaker and statistically non-significant as minority share increases. The diagnostic results, including heteroskedasticity, RESET test significance, residual tail deviations, and influential observations, indicate that the model is informative but not definitive. These limitations should be explicitly acknowledged in the results and discussion sections.

Pollution Exposure Clusters



The motivation for this analysis is to move beyond examining variable-by-variable relationships independently and instead try to understand how pollution variables operate as environmental stressors across Chicago neighborhoods. In urban settings, communities are not exposed to just a single pollutant. However, they are overburdened by a range of environmental pollutants like air pollution, traffic emissions, being close to hazardous sites, and industrial activity. Previous models have shown that treating those exposures separately can obscure underlying structural patterns of environmental inequality. For this purpose, a clustering approach was adopted to identify if Chicago communities naturally group into distinct environmental exposure profiles based on different elements of pollution. The environmental exposure clusters are reflective of meaningful patterns of cumulative environmental disadvantage across Chicago communities. This indicates that these environmental pollutants are not randomly distributed. The k-means clustering for this project grouped communities based on the following multiple pollution-related indicators:

These variables capture direct air pollution exposure and spatial proximity to environmental hazards. The horizontal axis (PC1) generally reflects overall pollution intensity, with communities further to the right tending to experience higher combined levels of environmental burden across multiple pollutants. The vertical axis (PC2) captures variation in the type or composition of pollution exposure, distinguishing between different exposure patterns such as traffic-dominated, industrial-dominated, or mixed-source environments. As a result, separation

between clusters reflects differences not only in how much pollution exists but also in the nature of that pollution.

The green cluster on the far right represents communities with the highest overall pollution burden, indicating areas with consistently elevated exposure. Meanwhile, the purple cluster is more centrally located and tightly grouped, suggesting moderate and relatively homogeneous exposure conditions. On the other hand, the red cluster overlaps with the other groups, indicating mixed exposure environments. It does not clearly fit in with a single dominant pattern. And lastly, the teal cluster is quite unique as it has a specialized exposure. It is likely driven by a specific pollution type like traffic proximity rather than overall intensity of the pollution. All in all, the clustering pattern is effective in categorizing the Chicago neighborhoods into distinct pollution environments. These environments evidently differ in structure and intensity.

Conclusion:

This study examined how environmental, socioeconomic, and demographic conditions are associated with respiratory health outcomes across Chicago neighborhoods using a multi-scale analytical design. Because the available datasets were measured at different geographic resolutions, the analysis used census block groups for the asthma model and census tracts for the COPD heat-health model. The findings suggest that asthma and COPD are both associated with neighborhood vulnerability, but the patterns differ by outcome and scale.

The asthma model shows that social and demographic factors are strongly associated with asthma prevalence at the census block group scale. The quadratic model provides a better fit than the interaction model, indicating that the relationship between minority population share and asthma prevalence is nonlinear rather than primarily conditional on educational attainment. Minority population share is positively and significantly associated with asthma prevalence, and the positive quadratic term suggests that this association becomes stronger at higher levels of minority population share. Older adult population share is also positively associated with asthma, while linguistic isolation is negatively and significantly associated with asthma in the model. Education-related variables show weaker and less consistent independent associations, and the interaction between minority share and educational attainment is not statistically significant. These results suggest that asthma prevalence is more consistently associated with demographic vulnerability, especially racial/ethnic composition and age structure, than with the tested minority-by-education interaction.

The COPD model shows a different pattern at the census tract scale. Summer maximum land surface temperature is positively associated with adult chronic obstructive pulmonary disease (COPD) prevalence, but this association varies by racial and ethnic minority share. The LST-COPD association is strongest in tracts with lower minority share and weakens as minority share increases. The Johnson–Neyman analysis indicates that the association is statistically significant only when minority share is below approximately 61%. This finding does not support the initial expectation that higher-minority tracts would show stronger heat-related COPD associations. However, it should not be interpreted as evidence that minority communities are less vulnerable to environmental health risks. A more cautious interpretation is that COPD prevalence in higher-minority or more disadvantaged tracts may already be shaped by other structural risks, including socioeconomic disadvantages, air pollution, housing conditions, healthcare access, or long-term

cumulative exposure. As a result, LST may add less marginal explanatory power once these factors are included in the model.

Taken together, the two models indicate that respiratory health vulnerability in Chicago is multidimensional and outcome-specific. Asthma appears to be more strongly associated with social and demographic vulnerability at the block group scale, while COPD shows a conditional heat-health association at the tract scale. These findings support the use of a multi-scale approach: block group analysis captures finer-grained neighborhood vulnerability, whereas tract-level modeling allows heat exposure and adult COPD estimates to be harmonized in a common analytic unit. However, because the two models operate at different spatial scales, coefficient magnitudes should not be directly compared. The results should instead be interpreted as complementary evidence of how environmental and social conditions structure respiratory health across Chicago neighborhoods.

Several limitations should be noted. Both models are cross-sectional, so the findings should be interpreted as associations rather than causal effects. The asthma and COPD outcomes are based on small-area estimates rather than individual clinical records, which may introduce spatial smoothing and measurement uncertainty. Model diagnostics also indicate heteroskedasticity, and the COPD model shows additional concerns, including a significant RESET test and influential observations. Future research should test the robustness of these findings using spatial regression, sensitivity checks, and additional variables such as housing quality, air conditioning access, smoking prevalence, tree canopy, healthcare access, and indoor environmental conditions.

The results suggest that public health and planning interventions should not rely on a single environmental exposure or a single demographic indicator to identify respiratory vulnerability. In Chicago, asthma and COPD are associated with different combinations of social vulnerability, demographic structure, air quality burden, and heat exposure. Effective neighborhood prioritization should therefore consider cumulative vulnerability, including baseline respiratory health burden, socioeconomic disadvantages, racialized environmental exposure, and local heat conditions. Such an approach can better support targeted interventions, including urban greening, heat mitigation, air pollution reduction, housing improvement, and respiratory health programs in communities facing overlapping risks.

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Appendix

Variable category	Code variable	Source / raw column	Analysis column	Definition / description	Unit / type	Source / notes
Geographic identifier	STATEFP	STATEFP	constant = 17	Illinois state FIPS code	integer	Always 17
Geographic identifier	COUNTYFP	COUNTYFP	constant = 31	Cook County FIPS code	integer	Always 31
Geographic identifier	TRACTCE	TRACTCE	derived in CT_ID	Census tract code	6-digit code	Used to construct tract ID
Geographic identifier	BLKGRPCE	BLKGRPCE	BG-level only	Census block group code	integer	Block group-level identifier
Geographic identifier	GEOID	GEOID	BG-level only	12-digit FIPS code combining state, county, tract, and block group	integer / string	Block group-level identifier
Geographic identifier	CT_ID	derived from GEOID	CT_ID	11-digit census tract FIPS code	string	First 11 digits of GEOID
Geographic identifier	CA_NUM	CA_NUM	CA_NUM	Chicago community area number	integer	Values 1–77
Geographic identifier	CA_TEXT	CA_TEXT	CA_TEXT	Chicago community area name	string	Example: Austin, Hyde Park

Geographic coordinate	lat	INTPTLAT	lat	Latitude of spatial unit internal point	decimal degrees	Used for spatial analysis
Geographic coordinate	lon	INTPTLON	lon	Longitude of spatial unit internal point	decimal degrees	Used for spatial analysis
Population / weight	pop	ACSTOTPOP	pop	Total population	count	Used as aggregation weight
Aggregation descriptor	n_bgs	derived	n_bgs	Number of block groups within each census tract	count	CT-level aggregation descriptor
Health outcome	asthma	asthmacrdp	asthma	Adult asthma crude prevalence	percent	CDC PLACES small-area estimate
Health outcome	copd	COPDcrdpre	copd	Adult chronic obstructive pulmonary disease crude prevalence	percent	CDC PLACES small-area estimate
Health outcome	chd	CHDcrdprev	chd	Coronary heart disease prevalence	percent	CDC PLACES small-area estimate
Health outcome	lbw	pctlbw	lbw	Low birth weight rate	percent	Vital statistics; weighted by births

Health outcome weight	Total_birt	Total_birt	Total_birt	Total live births	count	Used as weight for lbw
Land surface temperature	LST_min	MIN	LST_min	Minimum land surface temperature within spatial unit	°C	Landsat 8 Collection 2 Level-2 Surface Temperature
Land surface temperature	LST_max	MAX	LST_max	Maximum land surface temperature within spatial unit	°C	Primary heat exposure variable in main models
Land surface temperature	LST_mean	MEAN	LST_mean	Mean land surface temperature within spatial unit	°C	Derived from zonal statistics
Land surface temperature	LST_median	MEDIAN	LST_median	Median land surface temperature within spatial unit	°C	Derived from zonal statistics
Land surface temperature	LST_pct90	PCT90	LST_pct90	90th percentile land surface temperature	°C	Derived from zonal statistics
Land surface temperature	LST_within_sd	derived	LST_within_sd	Standard deviation of LST_max across block groups within each tract	°C	CT-level variability measure

Derived LST variable	LST_max_c	derived	LST_max_c	Mean-centered maximum LST	°C, centered	Used to reduce collinearity with squared term
Derived LST variable	LST_max_c2	derived	LST_max_c2	Squared centered maximum LST	squared °C	Quadratic term
Derived LST variable	LST_max_z	derived	LST_max_z	Standardized maximum LST	z-score	Used in interaction models
Derived LST variable	log_LST_max	derived	log_LST_max	Natural log of maximum LST	log scale	Used in elasticity/log models
Derived LST variable	LST_tertile	derived	LST_tertile	Categorical LST exposure group	low / mid / high	Based on quantiles
Demographic variable	minor	MINORPCT × 100	minor	Racial and ethnic minority population share	percent	EJSCREEN / ACS; CT-level values are 0–100
Demographic variable	lowinc	LOWINCPCT × 100	lowinc	Low-income population share	percent	EJSCREEN / ACS; CT-level values are 0–100
Demographic variable	lesshs	LESSHSPCT × 100	lesshs	Population share with less than high school education	percent	EJSCREEN / ACS

Demographic variable	lingiso	LINGISOPCT × 100	lingiso	Linguistically isolated households	percent	EJSCREEN / ACS
Demographic variable	under5	UNDER5PCT × 100	under5	Population under age 5	percent	EJSCREEN / ACS
Demographic variable	over64	OVER64PCT × 100	over64	Population over age 64	percent	EJSCREEN / ACS
Socioeconomic variable	unemp	unemprate	unemp	Unemployment rate	percent	ACS; already measured as 0–100
Socioeconomic variable	hsgburden	P_hsg × 100	hsgburden	Share of low-income households with housing costs greater than 30% of income	percent	HUD CHAS; economic stress indicator
Air pollution variable	pm25	PM25num	pm25	Annual PM2.5 concentration	µg/m ³	EJSCREEN
Air pollution variable	ozone	OZONEnum	ozone	Ozone concentration	ppb	EJSCREEN
Air pollution variable	dslpm	DSLPMnum	dslpm	Diesel particulate matter concentration	µg/m ³	EJSCREEN
Traffic exposure	ptraf	PTRAF	ptraf	Traffic density	vehicles/meter/day	EJSCREEN; highly right-skewed

Derived traffic variable	log_ptraf	derived from PTRAF	log_ptraf	Natural log of traffic density plus one	log scale	Preferred in regression because ptraf is right-skewed
Composite index	POLLUTBURD	POLLUTBURD	not retained	Pollution burden composite index	index score	Not used as predictor alongside constituent variables
Composite index	POPCHARWIT	POPCHARWIT	not retained	Population characteristics composite index	index score	Not used as predictor alongside constituent variables
Composite index	AQHINdexsc	AQHINdexsc	not retained	Air Quality Health Index score	index score	Not used as predictor alongside constituent variables
Composite index	AQHINdex_1	AQHINdex_1	not retained	AQHI percentile rank	percentile	Not used as predictor alongside constituent variables
Data quality flag	FLAG	FLAG	filtered out	Data quality exclusion flag	text	Block groups with missing components removed
Data quality flag	FLAGGED	FLAGGED	not retained	Caution flag for incomplete or excluded observations	yes / no	Used during cleaning

Excluded variable	PNPL	PNPL	not used	Proximity to National Priorities List sites	indicator / score	Excluded due to limited variability
Excluded variable	PRMP	PRMP	not used	Proximity to Risk Management Plan facilities	indicator / score	Excluded due to limited variability
Excluded variable	PTSDF	PTSDF	not used	Proximity to treatment, storage, and disposal facilities	indicator / score	Excluded due to limited variability
Excluded variable	CANCERnum	CANCERnum	not used	Air toxics cancer risk	risk estimate	Excluded to avoid overlap with pollution indicators
Excluded variable	RESPnum	RESPnum	not used	Respiratory hazard index	index	Excluded to avoid overlap with pollution indicators
Excluded variable	*ptile variables	percentile-rank columns	not used	Percentile ranks of original variables	percentile	Excluded to avoid collinearity with original variables
Excluded geographic variable	ALAND	ALAND	not used	Land area	area	Not used as predictor

Excluded geographic variable	AWATER	AWATER	not used	Water area	area	Not used as predictor
Model outcome group	Outcomes	cleaned variables	asthma, copd, chd, lbw	Dependent variables used across model comparisons	percent	Main health outcomes
Model exposure group	Primary exposure	cleaned / derived variables	LST_max, LST_max_z, LST_max_c, LST_max_c2	Main summer heat exposure variables	°C / z-score	LST_max is primary exposure
Model control group	Demographic controls	cleaned variables	over64, under5, minor, lingiso	Demographic vulnerability controls	percent	Used in main regression models
Model control group	Socioeconomic controls	cleaned variables	lowinc, lesshs, unemp, hsgburden	Socioeconomic vulnerability controls	percent	Used in main regression models
Model control group	Pollution controls	cleaned variables	pm25, ozone, dslpm, log_ptraf	Environmental exposure controls	mixed units	Used in main regression models
Model moderator group	Interaction moderators	cleaned variables	minor, lowinc, over64	Moderators tested in interaction models	percent	minor is primary moderator
Spatial analysis input	Spatial coordinates	INTPTLAT, INTPTLON	lat, lon	Coordinates used for spatial analysis	decimal degrees	Used for spatial weight inputs

Aggregation rule	Population-weighted variables	multiple BG-level variables	CT-level variables	Most variables aggregated from block groups to census tracts using population weights	weighted mean	ACSTOTPOP used as primary weight
Aggregation rule	Birth-weighted variable	pctlbw, Total_birt	lbw	Low birth weight aggregated using live births as weight	weighted mean	Total_birt used as weight
Aggregation rule	Population total	ACSTOTPOP	pop	Total tract population	sum	Sum of block group populations
Aggregation rule	Live births	Total_birt	Total_birt	Total tract live births	sum	Sum of block group births
Aggregation rule	Block group count	derived	n_bgs	Number of block groups in each census tract	count	CT-level descriptor
Aggregation rule	Within-tract LST variability	derived	LST_within_sd	Standard deviation of LST_max across block groups in each tract	°C	Unweighted standard deviation

```
#libraries required to run the data analysis activities.
library(tidyverse)
library(ggplot2)
library(dplyr)
library(foreign)
library(janitor)
```



```

library(gt)
library(modelsummary)
library(corrplot)
library(sandwich)
library(lmtest)
library(ggplot2)
library(car)
library(broom)

# Running the dataset
df = read.dbf("PPMA541FinalDataset.dbf")

# Checking all the variables
names(df)

# rows and columns
dim(df)

df2 <- df %>%
  select(
# # REMOVE GEOGRAPHIC META

    -STATEFP, -COUNTYFP, -TRACTCE, -BLKGRPCE,
    -NAMELSAD, -MTFCC, -FUNCSTAT,
    -ALAND, -AWATER, -INTPTLAT, -INTPTLON,
    -GEOID, -GEOID_Num, -GEOID_1,
    -Field, -census_blo,
    -citiesctpo,

    -OBJECTID_1, -OBJECTID_2,
    -NEAR_FID, -NEAR_DIST,
    -Shape_Leng, -Shape_Area,
    -ZONE_CODE, -COUNT,
    -AREA,

    -PM25numpct, -Ozonenumpc, -DSLPMnumpc,
    -CANCERnump, -RESPnumpct,
    -PTRAFpctil, -PRMPpctile, -PTSDFpctil, -PNPLpctile,

    -asthmacr_1, -COPDcrdp_1, -CHDcrdpr_1,
    -PCTLBWpcti, -UNDER5PCTp, -OVER64PCTp,
    -LOWINCPCTp, -MINORPCTpc, -LESSHSPCTp,
    -LINGISOP_1, -UNEMPRATEp, -P_hsgpctil,

    -AQHIndexsc, -sensavgw_1, -NOvulnerab,
    -AQHIndex_1, -pollUTbu_1, -POPCHARW_1
  )

```

```

# Renaming variables
df_clean <- df2 %>%
  rename(

    # Geographic IDs (KEEP)
    census_block_group = cbgrouptex,
    census_block_group_id = cbgroupnum,
    census_tract = ctlongchar,
    census_tract_fips = ctfips,

    community_area = CA_TEXT,
    community_area_id = CA_NUM,

    # Population / demographics
    population = ACSTOTPOP,
    minority_pct = MINORPCT,
    low_income_pct = LOWINCPCT,
    less_hs_pct = LESSHSPCT,
    linguistic_iso_pct = LINGISOPCT,
    under5_pct = UNDER5PCT,
    over64_pct = OVER64PCT,

    # Environmental exposure
    proximity_traffic = PTRAF,
    proximity_national_sites = PNPL,
    proximity_risk_sites = PRMP,
    proximity_waste_sites = PTSDF,

    # Pollution
    diesel_pm_num = DSLPMnum,
    cancer_risk_num = CANCERnum,
    respiratory_hazard_num = RESPnum,
    ozone_num = OZONEnum,
    pm25_num = PM25num,

    # Health outcomes
    asthma_perc = asthmacrdp,
    copd_perc = COPDcrdp,
    chd_perc = CHDcrdp,
    low_birthweight_perc = pctlbw,

    # Socioeconomic
    unemp_rate = unemprate,
    housing_burden_perc = P_hsg,
    total_births_perc = Total_birt
  )

glimpse(df_clean)

```

Summary table for the relevant variables

```
rq1_data <- df_clean %>%
  select(
    asthma_perc,
    copd_perc,
    minority_pct,
    low_income_pct,
    less_hs_pct,
    linguistic_iso_pct,
    under5_pct,
    over64_pct,
    proximity_traffic,
    proximity_risk_sites,
    pm25_num,
    ozone_num,
    diesel_pm_num
  )

# Summary stats
datasummary_skim(rq1_data)

tbl <- datasummary_skim(
  rq1_data,
  output = "gt"
) %>%
  tab_header(
    title = "Summary Statistics: Asthma, COPD, and Environmental Indicators"
  )

gtsave(tbl, "rq1_summary_table.png")

library(dplyr)
library(gt)

# Create correlation table
cor_table <- df_clean %>%
  select(
    asthma_perc,
    copd_perc,
    minority_pct,
    low_income_pct,
    less_hs_pct,
    pm25_num,
    ozone_num,
    proximity_traffic
  ) %>%
  cor(use = "complete.obs")

# Convert to data frame for gt
cor_df <- round(cor_table, 2) %>%
```

```

as.data.frame()

# Add row names as a column
cor_df <- tibble::rownames_to_column(cor_df, var = "Variable")

# Create gt table
gt_tbl <- gt(cor_df)

# Save as PNG
gtsave(gt_tbl, "correlation_table.png")

# 1. Create lookup table for community names
ca_names <- tibble(
  ca_num = 1:77,
  community_name = c(
    "Rogers Park", "West Ridge", "Uptown", "Lincoln Square", "North Center",
    "Lake View", "Lincoln Park", "Near North Side", "Edison Park", "Norwood Park",
    "Jefferson Park", "Forest Glen", "North Park", "Albany Park", "Portage Park",
    "Irving Park", "Dunning", "Montclare", "Belmont Cragin", "Hermosa",
    "Avondale", "Logan Square", "Humboldt Park", "West Town", "Austin",
    "West Garfield Park", "East Garfield Park", "Near West Side", "North Lawndale",
    "South Lawndale", "Lower West Side", "Loop", "Near South Side", "Armour Square",
    "Douglas", "Oakland", "Fuller Park", "Grand Boulevard", "Kenwood",
    "Washington Park", "Hyde Park", "Woodlawn", "South Shore", "Chatham",
    "Avalon Park", "South Chicago", "Burnside", "Calumet Heights", "Roseland",
    "Pullman", "South Deering", "East Side", "West Pullman", "Riverdale",
    "Hegewisch", "Garfield Ridge", "Archer Heights", "Brighton Park", "McKinley Park",
    "Bridgeport", "New City", "West Elsdon", "Gage Park", "Clearing",
    "West Lawn", "Chicago Lawn", "West Englewood", "Englewood", "Greater Grand Crossing",
    "Ashburn", "Auburn Gresham", "Beverly", "Washington Heights", "Mount Greenwood",
    "Morgan Park", "O'Hare", "Edgewater"
  )
)

# Join community names + prepare asthma data + plot top 10

top10_asthma_plot <- df_clean %>%
  group_by(community_area_id, community_area) %>%
  summarise(
    asthma = mean(asthma_perc, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  left_join(ca_names, by = c("community_area_id" = "ca_num")) %>%
  arrange(desc(asthma)) %>%
  slice_head(n = 10)

# Asthma plot
png("top10_asthma.png", width = 1200, height = 800, res = 150)

ggplot(top10_asthma_plot, aes(x = reorder(community_name, asthma), y = asthma)) +
  geom_col(fill = "lightblue") +

```

```

coord_flip() +
theme_minimal() +
labs(
  title = "Top 10 Chicago Communities by Asthma Prevalence",
  x = NULL,
  y = "Asthma Prevalence (%)"
)

dev.off()

```

And now for COPD (chronic pulmonary disease)

```

# Join community names + prepare COPD data + plot top 10

top10_copd_plot <- df_clean %>%
  group_by(community_area_id, community_area) %>%
  summarise(
    copd = mean(copd_perc, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  left_join(ca_names, by = c("community_area_id" = "ca_num")) %>%
  arrange(desc(copd)) %>%
  slice_head(n = 10)

png("top10_copd.png", width = 1200, height = 800, res = 150)

ggplot(top10_copd_plot, aes(x = reorder(community_name, copd), y = copd)) +
  geom_col(fill = "maroon") +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Top 10 Chicago Communities by COPD Prevalence",
    x = NULL,
    y = "COPD Prevalence (%)"
  )

dev.off()

```

—X—X—X—X—

K-Means Clustering

To discover the pollutants that may impact asthma

```

# isolating the environmental stressors
pollution_data <- df_clean %>%
  select(
    asthma_perc,
    pm25_num,

```

```

    diesel_pm_num,
    ozone_num,
    respiratory_hazard_num,
    cancer_risk_num,
    proximity_traffic,
    proximity_risk_sites,
    proximity_waste_sites,
    community_area_id
  )

# Scaling the variable
pollution_scaled <- pollution_data %>%
  mutate(across(-c(asthma_perc, community_area_id), scale))

set.seed(123)

kmeans_model <- kmeans(
  pollution_scaled %>% select(-asthma_perc, -community_area_id),
  centers = 4,
  nstart = 25
)

# factor cluster
pollution_scaled$cluster <- as.factor(kmeans_model$cluster)

# compare asthma rates across clusters
pollution_scaled %>%
  group_by(cluster) %>%
  summarise(
    mean_asthma = mean(asthma_perc),
    mean_pm25 = mean(pm25_num),
    mean_diesel = mean(diesel_pm_num),
    mean_ozone = mean(ozone_num)
  )

pca <- prcomp(pollution_scaled %>% select(pm25_num, diesel_pm_num, ozone_num,
  respiratory_hazard_num, cancer_risk_num),
  scale. = FALSE)

pca_df <- data.frame(
  PC1 = pca$x[,1],
  PC2 = pca$x[,2],
  cluster = pollution_scaled$cluster,
  asthma = pollution_scaled$asthma_perc
)

png("pollution_cluster.png", width = 1200, height = 800, res = 150)
ggplot(pca_df, aes(PC1, PC2, color = cluster)) +
  geom_point(alpha = 0.7) +
  labs(
    title = "Pollution Exposure Clusters (PCA Space)",
    x = "PC1",

```

```

  y = "PC2"
)

library(dplyr)
library(foreign)
library(modelsummary)
library(sandwich)
library(lmtest)
library(ggplot2)
library(car)
library(broom)

df <- read.dbf("~/Documents/Project Plenty/PPMA541FinalDataset.dbf")

df_cleaned <- df %>%
  select(

    # Geographic metadata
    -STATEFP, -COUNTYFP, -TRACTCE, -BLKGRPCE,
    -NAMELSAD, -MTFCC, -FUNCSTAT,
    -ALAND, -AWATER, -INTPTLAT, -INTPTLON,
    -GEOID, -GEOID_Num, -GEOID_1,
    -Field, -census_blo,
    -citiesctpo,

    # Spatial geometry variables
    -OBJECTID_1, -OBJECTID_2,
    -NEAR_FID, -NEAR_DIST,
    -Shape_Leng, -Shape_Area,
    -ZONE_CODE, -COUNT,
    -AREA,

    # Percentile variables
    -PM25numpct, -Ozonenumpc, -DSLPMnumpc,
    -CANCERnump, -RESPnumpct,
    -PTRAFpctil, -PRMPpctile,
    -PTSDFpctil, -PNPLpctile,

    # Duplicate health variables
    -asthmacr_1, -COPDcrdp_1, -CHDcrdpr_1,

    # Percentile demographic variables
    -PCTLBWpcti, -UNDER5PCTp,
    -OVER64PCTp, -LOWINCPCTp,

```

```

-MINORPCTpc, -LESSHSPCTp,
-LINGISOP_1, -UNEMPRATEp,
-P_hsgpctil,

# Composite index variables
-AQHINdexsc, -sensavgw_1,
-NOvulnerab, -AQHINdex_1,
-pollUTbu_1, -POPCHARW_1
)

#Rename variable

asthma_environmental_factors_dataset <- df_cleaned %>%
  rename(

# Geographic identifiers
census_block_group = cbgrouptex,
census_block_group_id = cbgroupnum,
census_tract = ctlongchar,
census_tract_fips = ctfips,
community_area = CA_TEXT,
community_area_id = CA_NUM,

# Population and demographic variables
population = ACSTOTPOP,
minority_pct = MINORPCT,
low_income_pct = LOWINCPCT,
less_hs_pct = LESSHSPCT,
linguistic_iso_pct = LINGISOPCT,
under5_pct = UNDER5PCT,
over64_pct = OVER64PCT,

# Environmental exposure variables
proximity_traffic = PTRAF,
proximity_national_sites = PNPL,
proximity_risk_sites = PRMP,
proximity_waste_sites = PTSDF,

# Pollution variables
diesel_pm_num = DSLPMnum,
cancer_risk_num = CANCERnum,
respiratory_hazard_num = RESPnum,
ozone_num = OZONEnum,
pm25_num = PM25num,

# Health outcome variables
asthma_perc = asthmacrdp,
copd_perc = COPDcrdp,
chd_perc = CHDcrdp,
low_birthweight_perc = pctlbw,

```



```

# Socioeconomic variables
unemp_rate = unemprate,
housing_burden_perc = P_hsg,
total_births_perc = Total_birt
)

#center variables

asthma_environmental_factors_dataset <- asthma_environmental_factors_dataset %>%
mutate(
  minority_pct_c =
    minority_pct - mean(minority_pct, na.rm = TRUE),

  less_hs_pct_c =
    less_hs_pct - mean(less_hs_pct, na.rm = TRUE),

  linguistic_iso_pct_c =
    linguistic_iso_pct - mean(linguistic_iso_pct, na.rm = TRUE),

  over64_pct_c =
    over64_pct - mean(over64_pct, na.rm = TRUE)
)

# Quadratic model

quadratic_model <- lm(
  log(asthma_perc) ~
    minority_pct_c +
    I(minority_pct_c^2) +
    less_hs_pct_c +
    linguistic_iso_pct_c +
    over64_pct_c,

  data = asthma_environmental_factors_dataset
)

summary(quadratic_model)

# Robust

coeftest(
  quadratic_model,
  vcov = vcovHC(quadratic_model, type = "HC1")
)

# Interaction

interaction_model <- lm(

```

```

log(asthma_perc) ~
  minority_pct_c * less_hs_pct_c +
  linguistic_iso_pct_c +
  over64_pct_c,

data = asthma_environmental_factors_dataset
)

summary(interaction_model)

coeftest(
  interaction_model,
  vcov = vcovHC(interaction_model, type = "HC1")
)

# Model comparison

modelssummary(
  list(
    "Quadratic Model" = quadratic_model,
    "Interaction Model" = interaction_model
  ),
  vcov = "HC1",
  stars = TRUE
)

# multicollinrariy check

vif(quadratic_model)
vif(interaction_model)

par(mfrow = c(2, 2))

plot(
  quadratic_model,
  pch = 19,
  cex = 0.6,
  main = "Quadratic Model Diagnostics"
)

resettest(quadratic_model)

bptest(quadratic_model)

```

```

asthma_environmental_factors_dataset <- asthma_environmental_factors_dataset %>%
  mutate(

    minority_pct_z =
      as.numeric(scale(minority_pct)),

    less_hs_pct_z =
      as.numeric(scale(less_hs_pct)),

    linguistic_iso_pct_z =
      as.numeric(scale(linguistic_iso_pct)),

    over64_pct_z =
      as.numeric(scale(over64_pct))
  )

standardized_model <- lm(
  log(asthma_perc) ~
    minority_pct_z +
    less_hs_pct_z +
    linguistic_iso_pct_z +
    over64_pct_z,

  data = asthma_environmental_factors_dataset
)

summary(standardized_model)

modelsummary(
  list(
    "Quadratic Model" = quadratic_model,
    "Interaction Model" = interaction_model,
    "Standardized Model" = standardized_model
  ),
  vcov = "HC1",
  stars = TRUE
)

ggplot(
  asthma_environmental_factors_dataset,

```

```

aes(
  x = minority_pct_c,
  y = log(asthma_perc)
) +
geom_point(
  color = "cyan",
  alpha = 0.5
) +
stat_smooth(
  method = "lm",
  formula = y ~ x + I(x^2),
  color = "red",
  linewidth = 1.2
) +
labs(
  title = "Quadratic Model of Logged Asthma Rate",
  x = "Centered Minority Percentage",
  y = "Logged Asthma Rate"
) +
theme_minimal()

```

X-----X-----X

This part of the Project had a slight technical issue which is why the R-code for that couldn't be rendered through Quarto. Hence, the codes were directly copied and paste from the script and onto a word file. Hence, we're pasting it over here. We apologize for that. Thank you!

X-----X-----X

```

required_pkgs <- c("dplyr", "ggplot2", "lmtest", "sandwich", "interactions", "car")
new_pkgs <- required_pkgs[!required_pkgs %in% installed.packages()[, "Package"]]
if (length(new_pkgs) > 0) install.packages(new_pkgs)
invisible(lapply(required_pkgs, library, character.only = TRUE))

# Load data
df <- read.csv("chicago_1st_health_censustract.csv", stringsAsFactors = FALSE)

cat("Census tract sample size:", nrow(df), "\n")

```

```

cat("Community Areas covered:", length(unique(df$CA_NUM)), "\n\n")

# Transform variables
df$LST_max_z <- as.numeric(scale(df$LST_max))
df$log_ptraf <- log(df$ptraf + 1)

# Fit OLS interaction model
best_model <- lm(
  copd ~ LST_max_z * minor +
  over64 + under5 + lingiso +
  lowinc + lesshs + unemp + hsgburden +
  pm25 + ozone + dslpm + log_ptraf,
  data = df
)

cat("\n===== OLS model summary =====", "\n")
print(summary(best_model))

# Robust inference
robust_results <- coefest(best_model, vcov = vcovHC(best_model, type = "HC3"))
cat("\n===== HC3 robust inference =====", "\n")
print(robust_results)

# Marginal LST effects by minority share
co <- summary(best_model)$coefficients
b_lst_main <- co["LST_max_z", "Estimate"]
b_lst_int <- co["LST_max_z:minor", "Estimate"]
sd_lst <- sd(df$LST_max, na.rm = TRUE)

cat("\n===== Marginal LST effects =====", "\n")
cat("Predicted change in COPD prevalence for a 1-SD increase in LST_max (~",
  round(sd_lst, 2), " °C):\n\n", sep = "")

for (m in c(20, 40, 60, 80)) {
  marg <- b_lst_main + b_lst_int * m
  cat(sprintf("minor = %2d%%: marginal LST effect = %+4f pp\n", m, marg))
}

# Johnson-Neyman analysis
jn <- johnson_neyman(best_model, pred = LST_max_z, modx = minor, alpha = 0.05)
cat("\n===== Johnson-Neyman bounds =====", "\n")
print(jn$bounds)

# Interaction plot
p_interaction <- interact_plot(
  best_model,
  pred = LST_max_z,
  modx = minor,
  modx.values = c(20, 50, 80),
  interval = TRUE,
  x.label = "Summer LST_max (z-score)",

```

```

y.label = "COPD prevalence (%)",
main.title = "LST x Minority share interaction on COPD",
legend.main = "Minority %"
)

ggsave("fig_best_interaction.png", p_interaction, width = 9, height = 6, dpi = 150)

# Johnson-Neyman plot
png("fig_best_johnson_neyman.png", width = 1000, height = 650, res = 130)
print(
  jn$plot +
    labs(
      title = "Johnson-Neyman region of significance",
      subtitle = "Conditional effect of LST_max_z across minority share"
    )
)
dev.off()

# Residual diagnostics
png("fig_best_diagnostics.png", width = 1200, height = 1000, res = 130)
par(mfrow = c(2, 2))
plot(best_model, which = 1:4)
dev.off()

# Diagnostic tests
cat("\n===== Diagnostic tests =====", "\n")

reset_p <- tryCatch(
  resettest(best_model, power = 2:3, type = "fitted")$p.value,
  error = function(e) NA
)
cat(sprintf("RESET test p-value: %.3e\n", reset_p))

bp_p <- tryCatch(bptest(best_model)$p.value, error = function(e) NA)
cat(sprintf("Breusch-Pagan p-value: %.3e\n", bp_p))

vifs <- tryCatch(
  car::vif(best_model, type = "predictor"),
  error = function(e) tryCatch(car::vif(best_model), error = function(e2) NULL)
)

cat("\n===== Variance Inflation Factors =====", "\n")
if (!is.null(vifs)) {
  if (is.data.frame(vifs)) {
    print(round(vifs[, "GVIF", drop = FALSE], 2))
    max_vif <- max(vifs$GVIF, na.rm = TRUE)
  } else if (is.matrix(vifs)) {
    print(round(vifs, 2))
    max_vif <- max(vifs[, "GVIF"], na.rm = TRUE)
  } else {
    print(round(vifs, 2))
  }
}

```

```

    max_vif <- max(vifs, na.rm = TRUE)
  }
  cat(sprintf("\nMaximum GVIF/VIF: %.2f\n", max_vif))
}

# Model fit
cat("\n===== Model fit =====", "\n")
cat(sprintf("N: %d\n", nobs(best_model)))
cat(sprintf("R-squared: %.4f\n", summary(best_model)$r.squared))
cat(sprintf("Adjusted R-squared: %.4f\n", summary(best_model)$adj.r.squared))
cat(sprintf("AIC: %.2f\n", AIC(best_model)))
cat(sprintf("BIC: %.2f\n", BIC(best_model)))

cat("\nAnalysis complete. Output files generated:\n")
cat("- fig_best_interaction.png\n")
cat("- fig_best_johnson_neyman.png\n")
cat("- fig_best_diagnostics.png\n")

```